

Fig. 5: Structure of a sensor network

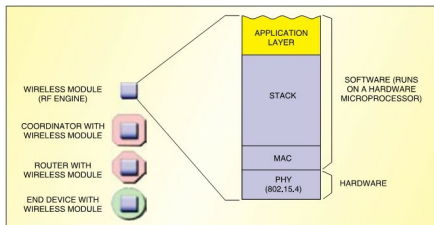


Fig. 6: The main elements of a wireless module

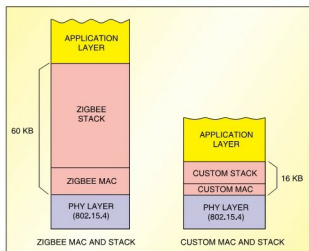


Fig. 7: Off-the-shelf vs custom MACs and stacks

So, while building a wireless sensor product that involves multiple layers of software, some factors need to be considered:

1. Is the code that ships with the board proprietary (incurring royalty), or is it open source?

2. Is the vendor planning to keep up with newer revisions of the standards?

3. What is the software roadmap of the vendor regarding integrated development environment (IDE), compilers, debug tools and real-time operating system (RTOS)?

4. Does the vendor offer some parts of its code as open source?

5. Can it be modified freely in the course of developing the product?

smaller application can do without an RTOS by using a loop with interrupt service routine (ISR). But for making bigger devices like routers and coordinators an RTOS is indispensable because it provides support to TCP/IP and USB mass storage devices, and helps in executing parallel software tasks. A version of Linux—uClinux—could be another choice. For small devices, a proprietary scheduler can do the task.

IDE and compilers. To select IDE and compiler, no rigid rules are applied as long as it can do the task well.

Debug tools. Debugging these devices while they are in operation is a difficult task because they are embedded and are in a network. A debugging agent can run on the device, communicating with the IDE running on the development PC, by adopting a 'step through the code' approach. But the problem with this approach is that the downloaded code has to contain this debugging agent and has to be synchronised with the debugger all the time. An alternative is to use a packet sniffing device and software for the wireless network under development.

Programming languages. Majority of the development processes use C/C++ as a development language. An interesting offer from Sun, called Sun SPOT, uses a special Java virtual machine (JVM) called Squawk. The development can be done in Java using popular IDEs like NetBeans to debug and deploy the application.

Elements of a wireless module

Each of the coordinators, repeaters/routers and sensor node units employ a wireless module comprising a sophisticated mixture of hardware and software to actually perform the RF communications, as shown in Fig. 6.

PHY, MAC and stack considerations. PHY is the lowest layer in the network, which transmits and receives

Royalties and source code. It is important to consider the royalties involved in the software layers before any decision is made because the network and transportation layers come in the form of a library, and how they can be used in a product is a major concern.

Compliance to standards. Some standards like ZigBee are updated almost every year, as

the compliancy of the solutions is very important. How compliant the solutions will be after one year is harder to gauge. One way to solve this problem is to find proof of certification for standards compliance.

Simple scheduler or RTOS. A